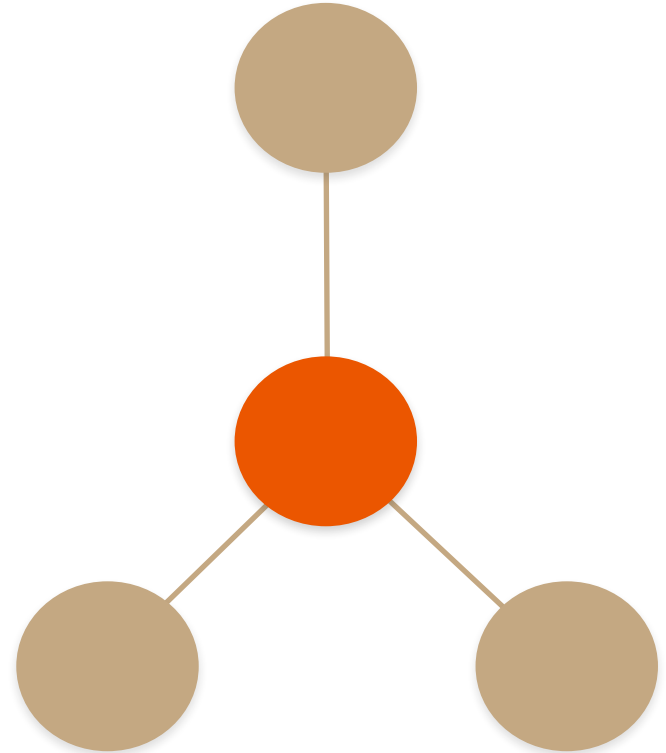


Dimensional Modelling - Lab

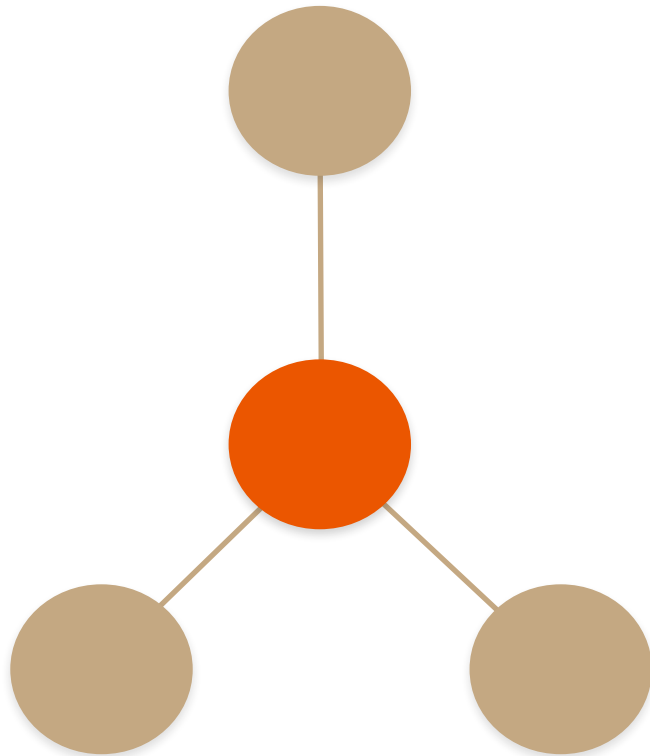
Lab - Building the Star Schema in SQL



Today's Lab Objectives

You designed it. Now you build it.

- **Write DDL for three dimensions and one fact table in PostgreSQL**
- **View the star shape as an ERD in DBeaver**
- **Load a seed dataset and run the first analytical query**
- **Deliverable: working star schema + ERD screenshot + one query result**



Recap: The Star We Designed



- Three dimensions: `dim_student`, `dim_course`, `dim_date`
- One fact table: `fact_enrollment`, referencing all three by foreign key
- Grain, stated exactly as the Theory deck left it:

One row in `fact_enrollment` represents one student enrolled in one course during one academic term.

DDL Walkthrough: dim_student

Surrogate key (SERIAL, warehouse-generated) plus the natural key carried over from OLTP.

```
CREATE TABLE dim_student (  
    student_key    SERIAL PRIMARY KEY,    -- surrogate key  
    student_id     INTEGER NOT NULL,      -- natural key from OLTP  
    first_name     VARCHAR(50),  
    last_name      VARCHAR(50),  
    email          VARCHAR(120),  
    date_of_birth  DATE,  
    nationality     VARCHAR(50)  
);
```

student_key is what every fact row will reference. student_id just traces a row back to its OLTP source - the fact never uses it directly.

DDL Walkthrough: dim_course

Same surrogate/natural key pattern - and department is denormalised onto every row.

```
CREATE TABLE dim_course (  
    course_key      SERIAL PRIMARY KEY,      -- surrogate key  
    course_id       INTEGER NOT NULL,        -- natural key from OLTP  
    course_name     VARCHAR(120),  
    department      VARCHAR(80),            -- denormalised onto every row  
    credits         INTEGER,  
    level           VARCHAR(20)  
);
```

department sits directly on dim_course - this is the deliberate denormalisation from the Theory deck, not an oversight.

DDL Walkthrough: dim_date

Built once, reused by every fact table in the spine - generated, not sourced from OLTP.

```
CREATE TABLE dim_date (  
    date_key          SERIAL PRIMARY KEY,  
    full_date         DATE NOT NULL,  
    semester          VARCHAR(20),  
    academic_year     VARCHAR(9),      -- e.g. '2023/2024'  
    month             INTEGER,  
    year              INTEGER,  
    quarter           INTEGER  
);
```

No natural key here because there is no OLTP "date" table - date_key is the only key this dimension needs.

DDL: The Fact Table

Three foreign keys, one row per student/course/term. Every key is NOT NULL.

```
CREATE TABLE fact_enrollment (  
    enrollment_key    SERIAL PRIMARY KEY,  
    student_key       INTEGER NOT NULL REFERENCES  
dim_student(student_key),  
    course_key        INTEGER NOT NULL REFERENCES dim_course(course_key),  
    date_key          INTEGER NOT NULL REFERENCES dim_date(date_key),  
    grade             VARCHAR(2),           -- outcome; NULL until term  
ends  
    enrollment_count  INTEGER NOT NULL DEFAULT 1    -- additive measure  
);
```

The surrogate-FK pattern: every FK points at a dimension's surrogate key, never its natural key. This is what draws the three spokes of the star.

Activity: Write the DDL

Create the three dimensions first, then the fact table with its foreign keys.

Run each CREATE TABLE statement and confirm it succeeds.

If a statement fails, read the error message before retrying.



40 min

View the ERD in DBeaver

- Open the Database Navigator and expand your schema
- Right-click the schema node -> View Diagram (or open the schema and select the ER Diagram tab)
- DBeaver Community auto-draws the diagram from your foreign keys - no extra setup needed
- Confirm the star shape: fact_enrollment sits centrally with one line to each of the three dimensions
- This diagram, saved or screenshotted, is your ERD deliverable

Seed Data: Loading by INSERT, for Now



- Today you load a small seed dataset directly with INSERT statements
- ETL from the real OLTP source begins Day 3 - today is about proving the schema works
- Dimension seed: 6 students, 4 courses, 4 terms (two academic years, two semesters each)
- Fact seed: about 22 enrolments, spread unevenly across the four terms
- Grade is filled in for the three past terms, and left NULL for the current term - the outcome is not yet known at enrolment

Activity: Load Seed Data + Query

Load the seed rows using the INSERT script in your notes.

Then run the enrollments-per-semester query.

Record the result table you get back.



40 min

The Analytical Query, Annotated



```
SELECT dd.semester, dd.academic_year,  
       COUNT(fe.enrollment_key) AS total_enrollments  
FROM fact_enrollment fe  
JOIN dim_date dd ON fe.date_key = dd.date_key  
GROUP BY dd.semester, dd.academic_year  
ORDER BY dd.academic_year, dd.semester;
```

Additive-measure equivalent: `SUM(fe.enrollment_count)` returns the same totals as `COUNT(fe.enrollment_key)`. This SUM pattern is the one that generalises to every additive measure you meet later.

Expected Result



semester	academic_year	total_enrollments
Semester 1	2023/2024	5
Semester 2	2023/2024	6
Semester 1	2024/2025	7
Semester 2	2024/2025	4

- **What correct looks like: a 4-row result, one row per term, and the counts genuinely differ between rows.**
- **If the counts look wrong: check for missing seed rows, a wrong join column, or a GROUP BY that omits a selected column.**

Validate Against Day 1 Requirements



- Does this three-dimension star answer the questions you sketched on Day 1?
- Directly answered: enrolment trends by term, by course, and by student demographic (nationality, etc.)
- Needs more: anything about who taught a course - that needs a fourth dimension we have not built yet
- This gap is exactly what the fast-finisher extension below adds

Today's Deliverable



- The running DDL: keep your SQL script or DBeaver SQL editor tab open
- The DBeaver ERD screenshot: capture the diagram showing the star shape
- The query result: copy or screenshot the 4-row output from the semester query
- Submit or photograph all three as instructed

Deliverable, verbatim: "Working star schema + ERD screenshot + one query result."

Fast Finisher (Optional): Add dim_instructor

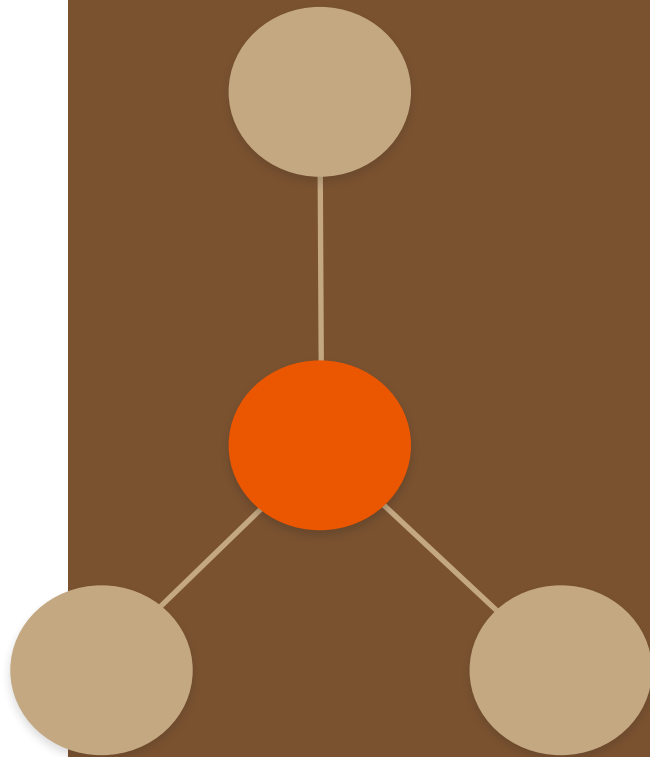
Optional fast-finisher extension: add a fourth dimension for per-lecturer analysis.

```
CREATE TABLE dim_instructor (  
    instructor_key    SERIAL PRIMARY KEY,  
    instructor_id     INTEGER NOT NULL,  
    first_name        VARCHAR(50),  
    last_name         VARCHAR(50),  
    department        VARCHAR(80)  
);  
  
ALTER TABLE fact_enrollment  
    ADD COLUMN instructor_key INTEGER REFERENCES dim_instructor(instructor_key);
```

Answers "which lecturers teach the highest-enrolment courses?" - beyond the core deliverable, so complete it only after the required schema, ERD, and query are done.

Questions & Discussion

Day 2 Lab - Building the Star Schema



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Last modified: 5 Jul 2026